

## Question 1 Highlighted Section

```
function[] = towers(n,frompeg,topeg,auxpeg);
    if (n == 1)
        fprintf('\t move disk 1 from peg %c to peg %c \n',frompeg,topeg);
    else
        towers(n-1, frompeg,auxpeg,topeg);
        fprintf('\t move disk %d from peg %c to peg %c \n', n, frompeg,topeg);
        towers(n-1,auxpeg,topeg,frompeg)
    end
end

clear all;
fprintf('Towers of Hanoi \n\n')
```

Towers of Hanoi

```
num = input('Towers of Hanoi: How many disks?\n ');
fprintf(' Moves required = %d \n', 2^num-1);
```

Moves required = 7

```
towers(num,'a','c','b');
```

```
move disk 1 from peg a to peg c
move disk 2 from peg a to peg b
move disk 1 from peg c to peg b
move disk 3 from peg a to peg c
move disk 1 from peg b to peg a
move disk 2 from peg b to peg c
move disk 1 from peg a to peg c
```

```
fprintf('\n\n');
```

## Q3 Highlighted Section

```
function generate_recursive(n, current)
    if length(current) == n
        if mod(sum(current == '0'), 2) == 0
            disp(current);
        end
        return;
    end
    for digit = '0':'9'
        generate_recursive(n, [current, digit]);
    end
end
```

```
n = input("Enter number: ")
```

```
n =
2
```

```
generate_recursive(n, '');
```

```
00
11
```

12  
13  
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```
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98
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```

## Q4 Highlighted section

```
n = input("Enter number: ")
```

```
n =
3
```

```
generate_recursive_base2(n, '');
```

```
111
110
101
011
010
```

```
function generate_recursive_base2(n, current)
    if length(current) == n
        disp(current);
        return;
    end
    generate_recursive_base2(n, [current, '1']);
    if isempty(current) || current(end) ~= '0'
        generate_recursive_base2(n, [current, '0']);
    end
end
```